

16 — Tech Stack & Dependency Justification

■ ■ Complete Technical Stack Table

System Layer	Technology / Library	Version	Why Chosen / Industry Advantage	Where Used in Project
Language	Python	3.12+	Universal language for data engineering, ML modeling, and web backend APIs	Core language across entire codebase
Backend Framework	FastAPI	0.128.8	High-performance asynchronous ASGI web framework with automatic Swagger docs	CODE_PH_0 & CODE_PH_1
ASGI Web Server	Uvicorn	0.39.0	Lightning-fast ASGI HTTP server implementation	Backend process listener (Port 8000)
Frontend UI	Streamlit	1.50.0	Python-native web application framework for rapid interactive dashboard rendering	CODE_PH_0 (Port 8501)
Charting Engine	Plotly	6.7.0	Dynamic, interactive JavaScript charts rendered seamlessly via Python	Forecast line graphs, radar charts, histograms
Data Manipulation	Pandas	2.3.3	Industry-standard library for time-series aggregation, filtering, and resampling	CODE_PH_0 , CODE_PH_1 , CODE_PH_2
Numerical Math	NumPy	2.0.2	High-speed array operations, linear algebra, and statistical sampling	ML regression feature matrix & Monte Carlo
Time-Series ML	Holt-Winters / Ridge	Custom	Exponential smoothing + L2 regularized regression for seasonal time-series	CODE_PH_0
Data Models	Pydantic	2.13.4	Strict data parsing, validation, and JSON schema serialization	FastAPI request/response validation
Containerization	Docker / Compose	2.x	Multi-container orchestration guaranteeing identical dev & prod execution	CODE_PH_0 , CODE_PH_1 , CODE_PH_2

■ Technology Choice Justification vs. Alternatives

1. FastAPI vs. Flask / Django

- **Why FastAPI?:** FastAPI is asynchronous by design, provides automatic interactive OpenAPI Swagger documentation at *CODE_PH_0*, and serializes Pydantic models up to 300% faster than traditional Flask.

2. Streamlit vs. React / Angular

- **Why Streamlit?:** Allows pure Python developers to build complex, responsive web UI layouts with interactive sliders, dataframes, and Plotly charts without needing Node.js or complex state-management boilerplate.

3. Holt-Winters + Ridge Regression vs. Deep Learning (LSTM / Prophet)

- **Why Holt-Winters + Ridge?:** Monthly port traffic datasets (1,354 records) are structured tabular time-series. Deep learning models like LSTM require millions of data points and tend to overfit small port datasets. Ridge Regression with L2 regularization guarantees stable, explainable weights with zero overfitting.